Password Cracking Detection and Prevention System for Login Page: Anomalous Device Detection and Response

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Declaration

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Abstract

The project addresses the critical issue of unauthorized email account access caused by weak or stolen passwords, which contribute to 81% of hacking-related breaches. Often, users are unaware of account takeovers until significant damage has occurred. To tackle this, the proposed solution, "Password Cracking Detection and Prevention System for Login Page with Anomalous Device Detection and Response," is designed to enhance security for both individual and organizational users. The system works by utilizing advanced real-time anomaly detection algorithms to identify suspicious login activities, particularly from unfamiliar devices. When such activities are detected, it immediately alerts users via email, providing actionable options to log out of all active sessions and reset their passwords promptly.

The solution is implemented by integrating with existing security measures, leveraging advanced algorithms, and focusing on usability to empower users to take swift action against potential threats. It will be deployed during login attempts, continuously monitoring for unusual patterns to ensure proactive prevention. The outcome is expected to include a substantial reduction in unauthorized access incidents, improved digital hygiene, and enhanced protection for sensitive personal and organizational data. The project ultimately combines technical innovation with user empowerment to create a robust and reliable security framework.

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Chapter 1

Introduction

1.1 Background and Context

In a way, cybersecurity has started getting embedded into many modern aspects of living today, which are globalizing online platform communication, work, financial transactions, and even for storing data. It forms perhaps one of the core most issues from cybersecurity, the protection of user credentials. Because login pages usually become those first gateways that essentially lead to sensitive information, the landscape of cybersecurity has grown big with an increase in remote working, e-commerce, and online services. Unarguably, robust security measures will be crucial in avoiding unauthorized access.

In the very beginning, computer networking was an emerging field and so was the equal threat to password security. As technology became better and advances were forged, it started proving the old ways of password-based authentication rather ineffective. It was only in the 90s, well into the early 2000s, that strong password policies were enforced along with compliance on password security. But as the threat of the internet grew and more data breaches were disclosed, came the introduction of methods like Two-Factor Authentication (2FA) and Multi-Factor Authentication (MFA) which were invented to counter both password theft and reproduction.

Weak and stolen passwords, credential stuffing, and phishing seem to be the most dangerous vectors at this stage. Indeed, it has been determined that 81% of hacking breaches involve weak or compromised passwords. More recently, bad actors have started using machine learning and AI in order to automate and accelerate traditional password-cracking techniques that defenses have become too good at defending against. Such a scenario will require adaptive, relevant solutions that respond in real time whenever unauthorized access attempts are made.

Today, the topic has gained even more critical relevance in the context of a rapidly changing digital environment where cyber-attacks are increasingly becoming more frequent and sophisticated. In addition to financial losses and data theft, unauthorized access to an email account or to other sensitive data has a really deep negative impact on the reputation of individuals and organizations.

Typically, as a consequence of such breaches, identity theft, leakage of confidential information, and often a long recovery phase follow.

Moreover, even traditional password protection methods have not enabled one to guard against such threats. Thus, enhancement of security lets anomaly detection and quick-response mechanisms reduce unauthorized access to a greater degree. Such measures could be used to detect suspicious activities at an early stage, for example, initial attempts from unknown devices or locations, giving warnings for users and thus enabling them to take immediate action. The approach productively closes this window of attack opportunity for attackers, builds trust among users, and further supplements better digital hygiene for personal and organizational information.

1.2 Problem Statement

To put it plainly, this project is more likely to provide solutions to unauthorized access into someone's email accounts using password cracking techniques. However, eyewitness reports suggest that many users still have weak passwords or reuse them. This has prompted the suggestion that international security improvements come with risks to users' account safety: Since most attack cybercriminals are intended to attack this weakness, there are various forms of attacks, for example, brute force attacks, dictionary attacks, and so many more that enable these unauthorized persons to access an individual's account without consent. Subsequently, the attackers can perform unauthorized transaction activities, change settings of the account, and also use the account for other illicit activities.

Their network might be broken down into micro-interference. Damage has already been done by the time the user realizes that their account has been hacked. The perpetrators are usually very subtle, so they can go on for very long periods without being discovered. Changes in security settings could involve completely locking out the legitimate user. It is compounded by reusing compromised credentials across a majority of services as this leads to increased damage because malicious actors can operate with a single stolen password to a wide assortment of accounts.

The importance of this problem is very wide because it refers to quite a large number of individuals and organizations. Most of the time, email addresses are the first port of call for stealing sensitive information, while also the main hub around which the rest of their online lives can be managed. And with such accounts compromised, not only private data can be lost but also financial records and potentially proprietary business information as well as access to critical services. Increasingly complex attacks that cyber-attackers use against passwords can easily be remediated for the adoption of more efficient security practices.

The consequences of account takeovers can be quite dire as they include financial loss, identity theft, disclosure of sensitive information, and a full house of description in the daily operation. Account breaches for organizations also come with reputational damage, regulatory fines, and loss of customer trust. Therefore, there is a necessity for advanced security measures which should be implemented as traditional password-only authentication has proven to be inadequate to curb account takeovers.

The proposed solution focuses on detecting unauthorized access of users thereby informing them of their anomalous device activity which is an essential step in closing this security gap. The alert notifies users of potentially fraudulent logins from a device that has not previously been used, enabling them to take immediate action such as logging out of all sessions and reset passwords before damage can be done. It has been designed proactively to mitigate risks caused by compromising accounts thus offering enhanced security for users and organizations and reducing the chance of widespread breaches.

1.3 Objectives and Project

1.3.1 Develop a Real Time Anomaly Detection System

The primary objective is to develop an anomaly detection system capable of monitoring login attempts in real-time. This system will leverage advanced algorithms to identify suspicious

activities, such as logins from unfamiliar devices or locations. By distinguishing legitimate user behavior from potential unauthorized access attempts, the system aims to provide proactive cybersecurity measures.

Unlike traditional security methods that rely heavily on static password restrictions or post-incident responses, this approach seeks to minimize the attack window by dynamically adapting to evolving threats. The result will be a real-time security framework that responds to anomalies as they arise, shifting the focus from reactive to adaptive cybersecurity.

1.3.2 Implement an Automated Notification System

Another key objective is to implement an automated notification system to promptly alert users via email when anomalous login attempts are detected. These alerts will include essential details, such as the unauthorized IP address and device information, enabling users to quickly assess the legitimacy of the login attempt.

This notification system enhances user awareness by informing account holders of potential malicious activities in near real-time. By empowering users to make timely and informed decisions, this feature mitigates risks associated with delayed breach detection and strengthens overall account security.

1.3.3 Deploy User-Initiated Security Mechanisms

The project also focuses on deploying user-initiated security mechanisms that allow users to take immediate action when anomalies are detected. These mechanisms include logging out of all active sessions, and prompting users to reset their passwords. By enabling users to secure their accounts proactively, the system strengthens account protection and minimizes the risk of further unauthorized access.

This approach promotes a more resilient, user-centric cybersecurity model by encouraging swift and decisive user actions. Beyond individual account security, these mechanisms contribute to broader cybersecurity practices by fostering active user involvement in threat mitigation.

1.4 Scope and Limitations

The project, "Password Cracking Detection and Prevention System with Anomalous Device Detection and Response", comprises the conception, realization, and testing of the complete security solution that is designed to fortify the protection of an email account against unauthorized access. This project base is on the following areas:

1. Anomaly Detection Mechanism: Designing an advanced algorithm that will monitor login patterns and detect suspicious or unfamiliar device logins as they happen.
2. Automated User Notification System: Create a complete email alert system to notify users in real-time whenever strange login events occur along with details about the accessing device.
3. User-Controlled Response Tools: Implementing the feature to users to log out all active sessions with prompts to change their password proactively in case of unauthorized access.
4. Device Tracking and Logging: Develop a system that maintains history records of devices commonly used to access the account so that the detection accuracy of anomalies login could be enhanced.

The project would be just on email account safety, majors in detection and user response rather than other mechanisms that strengthen passwords or integrated multi-factor authentication (MFA).

Therefore, this project is designed to significantly improve email account security, but there are some inherent limitations and constraints attached to it which may inhibit its implementation or outcome:

1. User Awareness Dependent: The system relies on users to act to the alerts it gives them. Any user who ignores messages would not recognize the importance of logging out and changing passwords to be susceptible to unauthorized access.

2. Scope of Detection: This project mainly concerns anomalous device logins, and it is probable that other sophisticated cyber attacks, such as phishing attacks or malware that evades conventional detection systems, would fall under its purview.
3. Resource and Infrastructure Constraints: Real-time detection and notification systems require considerable supporting infrastructure-including server resource and internet bandwidth-to be set up. This can therefore limit its feasibility for applications by organizations that do not have really good IT resources.
4. Privacy Concerns: The collection and retention of records about the user devices and the respective IP addresses may cause privacy issues. Those must be made sure that they are kept and managed in the manner prescribed by the data protection rules. Otherwise, that would further add to the complexity of the project.

Despite these limitations, the project aims to produce an efficient, effective, and easy use security enhancement tool for users following the applicant's apprehension of taking almost immediate control over one's account security. Design and implementation of this initiative can be done with a view of addressing these constraints while keeping improvements to the minimum in this regard.

1.5 Significance of Project

The project is quite relevant in today’s digital landscape where personal and professional communications are via email accounts. Statistics indicate that 81% of breaches by hackers are a result of weak or stolen passwords. Thus, it becomes extremely important to ensure the safety measures concerning email accounts. This project is important since it tackles the vulnerability left by most password-based authentication systems with real-time anomaly detection and intervention mechanisms that can be instigated by users themselves.

What this project does involves taking cybersecurity a step ahead compared to many other existing projects. These days, most cybersecurity methods adopt complicated passwords and, rarely, a Two-Factor Authentication (2FA) method. However, this project introduces yet another defense measure: the ability to detect and respond to potential risks before extensive damages occur. Therefore, the system improves on the general security of email accounts, ensuring that real-time alerts are sent upon detection of abnormal login patterns. Thus, it also instills confidence in the users towards protecting their data due to high security.

The potential benefits of this project extend to individuals, organizations, and the broader field of cybersecurity:

1. User Account Monitoring: Users will have more control over possible malicious login activity through email alerts immediately notifying the user about activity. This will also allow the user to take actions themselves, such as logging out from active sessions and changing passwords to avoid a long-term unauthorized access event. There will be more information protection through the prevention of any actual unauthorized access to sensitive details.
2. Reduced Data Breaches: Addition of real-time anomaly detection would also assist in significantly reducing successful password cracking and account taking. This makes it possible for early detection and intervention to stop attackers from pursuing more accounts or using one of those takeovers to get into other services or data infiltrations.
3. Encouragement of Proactive Cyber Hygiene: The focus on user awareness and proactive account management induces better digital habits. Users become more aware of their actions on accounts and therefore are encouraged to form a culture of cybersecurity awareness.
4. A Scalable Security Framework: This approach is flexible enough to act as a general framework for considering other online account protection systems as well. The principles of anomaly detection, user notification, and user-initiated security responses can apply far

beyond email accounts-for example to social media accounts, financial services, and corporate applications.

1.6 Organisation of Report

This report consisted of 7 chapters and the sequence is arranged as follows:

1. Chapter 1: Introduction

   This Chapter provides an overview of the project, including its background, objectives, significance, and scope. It sets the stage for the rest of the report by introducing the topic and outlining the key aspects of the project.

2. Chapter 2: Literature Review

   In this chapter, a comprehensive review of relevant literature and existing research in the field of encryption, data security, and related technologies is presented. The literature review serves to contextualize the project within existing knowledge and identify gaps and areas for further investigation

3. Chapter 3: Methodology and Requirements Analysis

   This chapter outlines the methodology employed in the project, including the research approach, data collection methods, and analysis techniques. It also provides a detailed analysis of the project requirements, including functional and non-functional requirements, to guide the system design and implementation process.

4. Chapter 4: System Design

   In this chapter, the design of the encryption platform is presented in detail. This includes the architectural design, database schema, user interface design, and other technical specifications. The system design chapter serves as a blueprint for the implementation phase of the project.

5. Chapter 5: Implementation and Testing

   This chapter focuses on the implementation of the encryption platform and the testing procedures employed to ensure its functionality, reliability, and security. It includes details of the development process, software tools used, and the results of various testing methods employed.

6. Chapter 6: System Deployment

   In this chapter, the process of deploying the encryption platform in a real-world environment is described. This includes considerations for deployment, configuration, and integration with existing systems or infrastructure. The system deployment chapter discusses the practical aspects of making the encryption platform operational.

7. Chapter 7: Discussions and Conclusion

   The final chapter of the report presents a discussion of the project findings, including an analysis of the results, implications for theory and practice, and recommendations for future research. It concludes with a summary of the key findings and contributions of the project, along with reflections on the overall project experience.

1.7 Summary

This chapter deals with the emerging importance of cybersecurity in user credentials. It highlights that the traditional password-based authentication technique is insufficient to stand against evolving cyber threats. Due to remote working, e-commerce, and other online services, there arises an increasing demand for the implementation of advanced security measures. The proposed project, "Password Cracking Detection and Prevention System with Anomalous Device Detection and Response," focus on detection and mitigation against unauthorized access to email accounts.

The project has highlighted the vulnerabilities due to weak passwords and the consequences brought about by account breaches, which include financial loss, identity theft, and reputational damages. The project intends to perform real-time anomaly detection that will pick up suspicious login attempts from unfamiliar devices or and alert the user through notifications, thus enabling them will tools for quick response through active sessions logging out and password resetting.

The key objectives are to design algorithms for anomaly detection, implement a notification system that provides real-time alerts, and enable security mechanisms initiated by the users. The project is limited to email account security and has various challenges like user awareness, resource availability, and privacy concerns. The project will contribute towards cybersecurity by proactively averting potential breaches.

The project is important because it fills in the gaps of traditional security methods and can provide a scalable, user-centered framework that can be expanded into other online accounts. This highlights early detection, user empowerment, and proactive cyber hygiene that will contribute to the greater field of cybersecurity and the reduction of risks of data breaches.

Chapter 2

Literature Review

Introduction

The literature review is supposed to be a critical review of the existing body of knowledge on the project, ensuring that the proposed solution is informed by a sound foundation of relevant theories, concepts, and existing technologies. This section helps identify best practices, understand strengths and limitations of current systems, and uncover gaps that the proposed project aims to fill. The literature review will be an important part of proving the project's relevance, as well as a theoretical and practical base for its elaboration.

2.1 Review of Relevant Theories and Concepts

2.1.1 Password Security and User Authentication

Theories and concepts related to password security and user authentication are fundamental in the field of cybersecurity. The password security and user authentication base is very important in the field of cybersecurity (Antonenko, 2022). Password-based authentication systems, being widespread, are naturally prone to various forms of attack vectors (Editor, 2022). Amongst these, techniques include brute force attacks, which test every possible combination systematically for a password, and dictionary attacks, precompiled lists used for efficient breaking of the system (Vigliarolo, 2019). The susceptibility of these systems is often compounded by user practices, such as weak password selection or reuse across platforms (Inc, 2024). Advanced methods include hash cracking, which exploits vulnerabilities in password storage mechanisms, highlighting the importance of robust hashing algorithms and multi-factor authentication as key defenses (Cypress Data Defense, 2020).

Understanding these weaknesses highlights the need for effective countermeasures, which this research seeks to address. The project focuses on implementing stronger authentication mechanisms through Multi-Factor Authentication (MFA) and advanced password hashing techniques. These methods not only minimize the risk of unauthorized access but also provide a solid foundation for effective breach identification and response mechanisms.

2.1.2 Anomaly Detection Algorithms

Anomaly detection algorithms are vital for identifying deviations from expected patterns, serving as key components in monitoring and securing systems (Bernard & Stryker, 2023). These algorithms use machine learning and statistical models to detect anomalies that might indicate security breaches or system faults (Coursera Staff, 2024).

The research leverages machine learning models, including clustering algorithms, rule-based systems, and supervised learning techniques such as neural networks and support vector machines, to develop adaptable and precise detection mechanisms (Quiroz-Vázquez, 2024). By training models on labeled datasets, these techniques enable the classification of events as normal or anomalous.

The research integrates these advanced methods into the system design to ensure resilience against emerging security challenges. By enabling the system to identify threat patterns and respond promptly, these mechanisms enhance the precision and flexibility of intrusion detection systems (Nailman, 2024).

2.1.3 User Awareness

User awareness is critical to effective cybersecurity strategies. The human factor is normally the weakest link, where a user may compromise security by being either negligent or lack of awareness (Leal, 2022). Research on the psychological and behavioral dimensions of cybersecurity highlights the importance of user engagement and education (Grobler et al., 2021). Research has shown that users are more responsive to alerts and notifications when they are short, contextually relevant, and accompanied by actionable advice (Team, 2022). Proactive measures include regular security training and the promotion of best practices that put users in a better position to recognize and mitigate potential threats (Ralph, 2023). The culture of security awareness will help an organization go a long way in improving its overall security posture.

The project incorporates these insights to develop a culture of security awareness. By increasing engagement and empowering users to take quick protective actions, such as responding to clear and actionable alerts, the measures reinforce technical safeguards. Ultimately, these strategies help organizations improve their overall security posture while supporting users in recognizing and mitigating threats effectively.

2.2 Study of Similar Existing Systems

2.2.1 Google

A review of existing systems, products, or solutions highlights Google has implemented a multi-layer security system that integrates advanced features such as anomaly detection and device activity monitoring (Google Account, 2024). It is great for sending notifications of suspicious activities in real time to users so that they can take immediate action to secure their accounts (Google Cloud, 2024). Moreover, the design by Google is intuitive and hence accessible to a wide variety of users. One advantage is that sometimes it generates false positives, which could be frustrating for users and reduce their confidence in the system (Google Design, 2024).

2.2.2 Microsoft

Microsoft focus is mostly on enterprise-level security, where the detection of suspicious sign-in attempts and delivery of detailed alerts to the administrator and user are emphasized (softchris, 2024). The system is designed to allow organizations to track and analyze potential threats comprehensively through its strong reporting capabilities (kimkischel, 2022). However, Microsoft's approach falls short in allowing users to take direct control over security incidents, relying more on administrative oversight than empowering individual users. (timurengin, 2023).

2.2.3 Mobile Legends

Within other fields, Mobile Legends offers a security feature that allows users to log out of other devices remotely (Jabel, 2023). This feature is convenient and effective for users in terms of

managing account access and preventing the misuse of their account, especially game assets. This feature, though crucial for securing gaming accounts, has a narrow scope when compared to more extensive systems such as Google and Microsoft. Security in Mobile Legends is designed to protect in-game, game-specific digital assets, not to provide broader cybersecurity (Mobile Legend, 2024).

2.2.4 Comparison Study

Features	**Google**	**Microsoft**	**Mobile Legends**	**Proposed System**
Target Users	General users	Enterprise environments	Gamers	General users
Strengths	User-friendly design, intuitive notifications, integrates security alerts	Advanced anomaly detection, detailed reporting for enterprise user	Easy account control with remote logout capability	User-friendly interface with anomaly detection, real-time alerts, and Multi-Factor Authentication (MFA)
Weaknesses	Generates false	Focused on	Lacks advanced	User awareness

	alarms, limited immediate options for reacting to events	enterprises, limiting its adaptability for personal account management	anomaly detection and real-time alerts	dependent, resource and infrastructure constraints
Anomaly Detection	Basic detection	Advanced detection	Lack detection	Advanced detection
Real-Time Alerts	Yes	Yes	No	Yes
Proactive Security	Limited	Yes	Minimal	Yes

Table 2.1: Comparison Study

A comparison of features, functionalities, strengths, and weaknesses across security systems reveals differing approaches based on target users. Google is strong in user-friendly design and integrates security alerts perfectly. It provides intuitive notifications that are accessible to the average user, which boosts the usability of its security. However, the system it has occasionally generates false alarms, which can frustrate users, while immediate options for the user to react to an event is limited (Thacker, 2018).

In particular, Microsoft has excelled with top-notch enterprise-level security-anomaly detection and detailed reporting, thus serving enterprise environments where detailed threat analysis by administrative personnel is a common case (yelevin et al., 2024). However, this does limit the adaptability of Microsoft in personal account management because of its focus on enterprise usage.

Mobile Legends has an easy, straightforward approach to account security; it allows users to manage access by remotely logging out of other devices (Jabel, 2023). This feature empowers users to maintain control over their accounts easily. However, advanced anomaly detection and

proactive real-time alerts are lacking in its security features. Furthermore, its focus is primarily on gaming accounts, which limits its application for broader digital protection (Synn, 2024).

An analysis of user feedback, adoption rates, and success stories reveals varied responses to security features across platforms. For instance, Mobile Legends' feature of logging out remotely has been positively received by users who value the control it provides over their account security (HotdogsHot, 2018). However, feedback often highlights the desire for more comprehensive measures such as real-time notifications for unusual activities or stronger proactive defenses seen in systems like Google's and Microsoft's.

2.3 Identification of Gaps in the Literature

Previous literature review has identified a few limitations in state-of-the-art security systems. One significant limitation is the lack of a good user-initiated response system. Most of the available systems effectively alert users to potential security threats, but they fail to provide immediate countermeasures, such as being able to log out from all active sessions. Another gap is in regard to balancing accuracy and usability. Current solutions often fail to provide accurate anomaly detection with low false positives, which can destroy user trust and satisfaction. Another limitation of systems is related to the integration of real-time threat detection with proactive user-initiated countermeasures, especially in mobile application environments. These gaps present opportunities for the development of more comprehensive and user-focused security frameworks.

In contrast, the proposed project attempts to bridge these gaps by incorporating a unified system that detects unauthorized logins and enables users to act in real time. Such real-time notifications together with user-controlled security responses are combined to provide better protection of accounts.

2.4 Key Findings

The literature review highlighted critical challenges and gaps in existing security systems, particularly in enhancing password security, improving anomaly detection algorithms, and increasing user awareness. Password-based authentication systems are vulnerable to brute force attacks, dictionary-based attacks, hash cracking, and weak password practices by users.
Addressing these vulnerabilities necessitates the adoption of strong multi-factor authentication (MFA) and advanced hashing techniques to provide a more secure foundation for user authentication.

Anomaly detection algorithms play a pivotal role in identifying security breaches by analyzing patterns and detecting irregularities using machine learning and statistical methods. While supervised learning techniques, such as neural networks, enhance detection accuracy, the trade-off between precision and usability remains an unresolved challenge. Real-time detection capabilities, combined with user-initiated response options, are especially important in mobile applications to empower users to quickly respond to potential threats.

User awareness is also a key factor in cybersecurity, as the human element is often considered the weakest link in any security setup. Engaging users through actionable alerts, training features, and user-friendly interfaces can significantly reduce risks and improve overall system security.
Comparative analyses of existing systems, such as those from Google, Microsoft, and Mobile Legends, reveal various strengths and weaknesses, while identifying significant gaps in user empowerment and the balance between accuracy and usability.

Building on these findings, this project seeks to address the identified gaps by integrating robust MFA mechanisms and enhanced password hashing techniques to strengthen authentication. Advanced anomaly detection algorithms will be developed using machine learning approaches such as clustering, rule-based systems, and supervised learning models to provide optimal threat detection with a minimal false positive rate. This balanced approach aims to enhance accuracy and usability, fostering user trust and improving system effectiveness.

Empowering users is at the core of the project, addressing existing limitations by incorporating real-time notifications and enabling user-initiated security actions. Features like logging out from all active sessions or temporarily locking accounts provide users with greater control over their security. The project also emphasizes user-centric design, featuring intuitive interfaces, actionable alerts, and security awareness programs to bolster the overall system.

By synthesizing insights from the literature, this project adopts a multi-layered security strategy that combines technical innovation with user-centric design. It addresses both technical vulnerabilities and behavioral factors, delivering an effective and comprehensive solution that balances security, usability, and user empowerment.

2.5 Summary

Chapter 2 discusses some of the current challenges in cybersecurity, such as password security, anomaly detection, and user awareness. This chapter highlights vulnerabilities of password-based systems, such as brute force and dictionary attacks, which are worsened by weak user practices. It emphasizes the need for robust multi-factor authentication and advanced password hashing as key defenses.

Anomaly detection algorithms are identified as key to detecting system irregularities and potential breaches. However, there remains the challenge of striking a balance between accuracy in detection and usability because the systems continue to generate false positives, which erode trust by users. The need for real-time detection integrated with user-initiated responses is particularly highlighted for mobile applications.

User awareness becomes critical security, where actionable alerts and training programs have turned out to be effective risk mitigants for human error. An analysis of existing systems, such as Google, Microsoft, and Mobile Legends, were analyzed based on various parameters like availability of features, user's empowerment, accuracy, and features related to mobile security to identify gaps.

The project aims to address these gaps by developing a unified security framework that will include robust authentication mechanisms, advanced anomaly detection algorithms, and real-time user-driven countermeasures.

Chapter 3

Methodology and Requirements Analysis

3.1 Project Methodology

The development methodology to be followed for the proposed Password Cracking Detection and Prevention System is Agile methodology, an iterative approach widely followed that focuses on collaboration, adaptability, and incremental progress.

The features of an Agile methodology for project development include a structure that emphasizes efficiency and quick adaptation to changing environments. It adopts iterative development in which the project is segmented into a series of short, convenient cycles, or sprints, with each sprint dedicated to realizing an operational advance of the system. Another feature is collaboration, involving an engaged role by active stakeholders, developers, and testers to produce a system meeting user needs and expectations. Besides, Agile also provides flexibility in incorporating modification based on feedback or a change in requirement to keep the product relevant and effective. Lastly, continuous testing and integration plays a crucial role in ensuring quality since it provides opportunities for testing frequently throughout development, locating issues at the earliest opportunity, and makes sure that a product remains usable.



Figure 3.1 Agile Project Development Diagram (Schnell, 2022)

The justification for selecting the Agile methodology is rooted in the project's requirements and constraints, which make it particularly suitable for development. With so many complex components like real-time anomaly detection algorithms through to user-initiated security activities, the iterative testing and refinement central to Agile ensure the optimization of such features in an effective manner. Agile takes a user-centric approach so that feedback is regular, ensuring that critical aspects such as alerting and response mechanisms are intuitive and user-friendly. Furthermore, the flexibility of the methodology is crucial in this dynamic cybersecurity environment, whereby new threats and changing user expectations require the embedding of emerging insights and improvements during the development cycle. By emphasizing incremental delivery, Agile also reduces risks: the ability to find and fix issues in areas like detection algorithms and alerting system integrations early on. Additionally, team collaboration promotes effective communication between developers, testers, and stakeholders-which is a necessity in order to implement multi-faceted functionalities for the system.

3.2 Requirements Gathering

The Password Cracking Detection and Prevention System with Anomalous Device Detection and Response will be a comprehensive security suite that will protect user accounts through the identification and mitigation of unauthorized access attempts in real-time. It takes up a structured approach to requirements gathering, integrating industry standards, regulatory frameworks, and user experience principles in the development of a robust, user-centric system.

Regulatory compliance is one of the cornerstones in the design of the system, guaranteeing compliance with international data protection standards such as the General Data Protection Regulation and the California Consumer Privacy Act. The system therefore gives priority to user privacy and acts within the boundaries of the law, thus building confidence in data handling and ensuring responsibility. It reduces legal risks and secures transparency and safety for users when their data is in process.

To further enhance the security, the system also integrates leading authentication standards such as OAuth, OpenID Connect, and FIDO (Fast Identity Online). These are highly recognized protocols for securing authentication workflows. Their integration strengthens the system by providing seamless compatibility with existing applications while facilitating secure login processes. Real-time anomaly-based checks are also implemented to detect suspicious activities on a proactive basis, thus assuring timely identification and response against potential threats.

The design follows the best practices of security as outlined by organizations such as OWASP (Open Worldwide Application Security Project). These include strong encryption, secure session management, and protection against common threats such as brute-force attacks and phishing attempts. By following these guidelines, the system reduces vulnerabilities significantly, hence offering a very secure environment for users.

Besides that, there is much usability focus. The system focuses on usability, having clear and actionable notifications, intuitive interfaces, and straightforward workflows that minimize user friction. This will enable or encourage proactivity in engaging security with the least possible disturbance of the experience.

Finally, the system design is informed by insights from successful implementations by leading organizations such as Google, Microsoft, and Mobile Legends. Key lessons learned include the importance of features like real-time anomaly detection, instant notifications, and user-initiated security actions. These practical and proven strategies ensure that the system is not only effective but also adaptive to changing security challenges.

3.2.1 Functional Requirements

The system incorporates a range of core functionalities designed to effectively achieve its security objectives. This consists of the core of an anomaly detection system that runs real-time monitoring of login, utilizing key parameters such as device information, geolocation, and user behavior

patterns. Such sophistication within the approach enables the system to identify and flag suspicious activities in real time to forestall unauthorized access.

Complementing the detection mechanism is a strong notification system that promptly alerts users it detects something wrong. The notifications will contain such critical information such as IP addresses, device types, and timestamps, meaning the users will have the necessary information to evaluate whether such activity is legitimate or not. Furthermore, the system enables users to take immediate, concrete steps directly from the notifications such as logging out from all devices or resetting passwords to minimize potential risks as fast as possible.

To enhance user control, the system features device management capabilities, thus enabling users to maintain a list of trusted devices. This list can be updated manually to ensure that only the authorized devices retain access to the account. This added layer of security empowers users to actively manage and safeguard their accounts.

For administrators, the system provides advanced analytics and reporting tools that show a complete picture of suspicious activities on the platform. These tools facilitate the monitoring and analyzing trends, resolving false positives, and implementing enhanced security measures where necessary.

These functionalities create a cohesive and powerful security framework that balances proactive threat detection, user empowerment, and administrative oversight in safeguarding accounts effectively.

3.2.2 Non-Functional Requirements

The system's non-functional requirements are designed to ensure scalability, reliability, and a user-centric experience. The identified requirements also form the basis for standards needed for a high-performance, secure, and accessible solution.

Performance and Responsiveness are critical priorities. The system must be able to detect anomalies within 5 seconds and notify the users, thus providing real-time feedback to reduce the risk of potential incidents. It is designed to handle up to 10,000 concurrent users without any degradation in performance, ensuring smooth operation even at peak usage.

Data security is a paramount in the design of the system, highlighting the principles of confidentiality, integrity, and availability. Confidentiality is ensured through the implementation of AES-256 encryption, widely recognized as one of the most secure encryption standards, to safeguard all sensitive information from unauthorized access. Integrity is maintained by employing TLS (Transport Layer Security) protocols to secure data transmission, preventing data tampering or interception during transit. To uphold availability, robust system architecture and failover mechanisms are incorporated to ensure that authorized users have uninterrupted access to their data and services, even in the event of unexpected disruptions. This comprehensive approach ensures the security of the system while fostering trust and reliability for its users.

The system is designed for 99.9% uptime to ensure uninterrupted service, meeting high-availability standards. This should help gain and retain the trust of the users as they would experience consistent protection against threats.

Usability is another core component of the system, from intuitive user interfaces serving both nono-technical to very technical people, ensuring ease in using and navigating the platform by all users.

By further consequence, the system will be highly maintainable because of its modular architecture, combined with following the best principles of clean coding. These design principles will enable for seamless updates, quick resolution of potential issues, and the effortless integration of new features to address emerging security challenges.

3.3 Requirements Analysis

3.3.1 Analysis and Prioritization

In the development of a Multi-Factor Authentication (MFA) system for the proposed platform, careful analysis and prioritization of requirements are essential to ensure a balanced integration of security and usability. Each requirement is analyzed based on how it impacts the security framework of the system and the improvement of the user experience.

The prioritization begins with the evaluation of the security importance of each feature. For example, the ability to let users log in securely using a combination of passwords and OTP verification is a core requirement that strengthens access control by preventing unauthorized access. Similarly, anomaly detection and alerts on suspicious activities are high-priority features that directly mitigate potential security breaches. Secure password reset mechanisms and session management, including across-device logout, are equally important in the provisions of addressing critical vulnerabilities.

Simultaneously, the features are considered regarding their usability impact to keep the system user-friendly. Complex security measures such as frequent OTP verifications may enhance the security but can frustrate the users. To address this, the system emphasizes flexibility through multiple authentication options like recognition of trusted devices and fallback methods to ensure that usability remains intact without compromising security.

The prioritization strategy focuses on the implementation of essential security measures first, such as robust anomaly detection, audit logging, and session management. Once these critical security features have been set up, a development team can focus on usability enhancements that include intuitively designed user interfaces for trusted device management, recovery processes, and notifications that are both clear and actionable. In the process, it balances security and usability to provide a safe, seamless user experience on the platform.

3.3.2 Identification of Dependencies and Conflicts Among Requirements

The integration of MFA into the proposed system has dependencies on various authentication factors and functionalities that need to be identified with care for smooth implementation.

Password Authentication requires secure storage mechanisms through hashing and salting to prevent vulnerabilities such as brute-force attacks or credential leaks. The implementation of password complexity policy influences user experience, while too restrictive requirements may lead to users getting frustrated and increasing support requests. OTP Verification involves dependencies on third-party services, such as email or SMS gateways, to deliver one-time codes.

Robust error-handling mechanisms and fallback options, such as resending OTPs or allowing alternative methods, are necessary to address issues like delivery failures or network outages.

Anomaly Detection would hence depend on analysis of login patterns, device metadata, and IP addresses for the detection of suspicious activities. This feature requires integration with analytics tools and databases capable of storing and processing behavioral data. Conflicts arise when false positives will create inconvenience for legitimate users, which requires fine-tuning the detection threshold and providing a feedback mechanism for users. The Management of Trusted Devices is dependent on securely storing device identifiers and maintaining a user-friendly interface for device management. The challenges are going to be in the compatibility of a wide variety of devices and edge cases, such as shared or public devices. Biometric Authentication is an option but relies on compatible hardware, such as fingerprint scanners or facial recognition cameras. Besides that, there are some problems that need to be considered, related to user consent, privacy policies, and device compatibility issues.

3.3.2.1 Minimize The Dependencies and Conflicts

To minimize these dependencies and conflicts, the system will adopt several strategies:

1. Compatibility Testing: Detailed assessment will be made at all levels of hardware, software, and third-party integrations to ensure flawless compatibility.
2. User Guidance: Provide clear instructions and support resources, such as tutorials and tooltips, to assist users in navigating new authentication features.
3. Alternative Methods: Offer fallback options like password-based login for users who face challenges with OTP or biometric authentication.
4. Dynamic Threshold Adjustment: Continuously refine anomaly detection algorithms to reduce false positives and enhance the accuracy of alerts.

By identifying and addressing dependencies early, the system can effectively integrate MFA features while maintaining a high standard of security and usability, delivering a seamless and secure user experience.

Chapter 4

System Design

System Design

In this chapter, detailed system design of the proposed system will be illustrated and described using various diagrams such as flowchart diagram, use case diagram, activity diagram, sequence diagram, class diagram, and system layouts.

4.1 Architecture Design

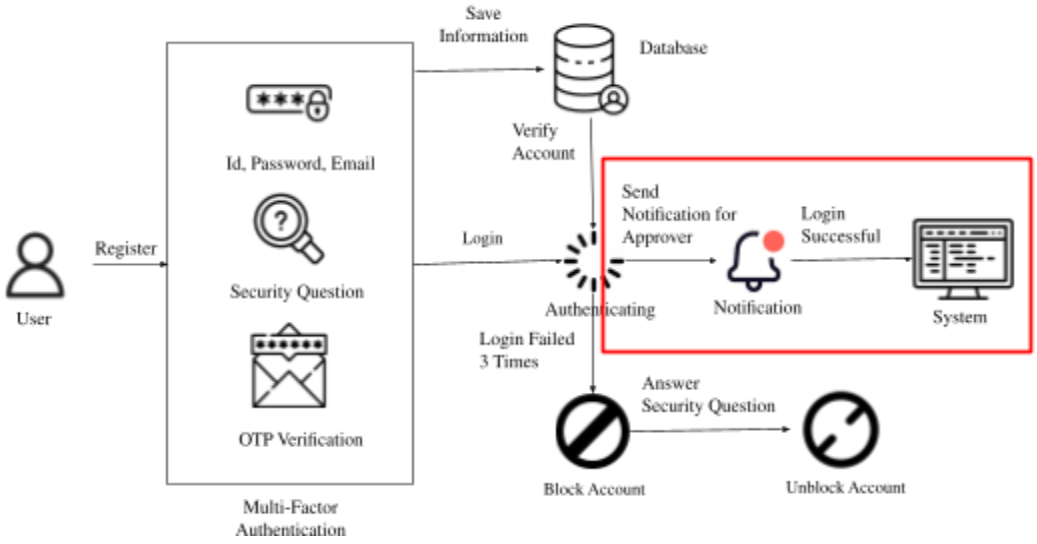


Figure 4.1: Concept Flow of The System

Figure 4.1 shows the basis concept flow of the proposed system. The red color box is mentioned for my proposed system. First, the users register their account by providing their ID, password, email, and security question. Then an OTP is sent to the user for verification after he/she has registered. The information will be stored securely in a database. During login, users provide their credentials, and the system verifies whether the user is registered or not. If yes, then the user will login successfully. If not, the system will send notification approver to the user email. After the user approves, then the user can login successfully.

4.1.1 Flowchart Diagram

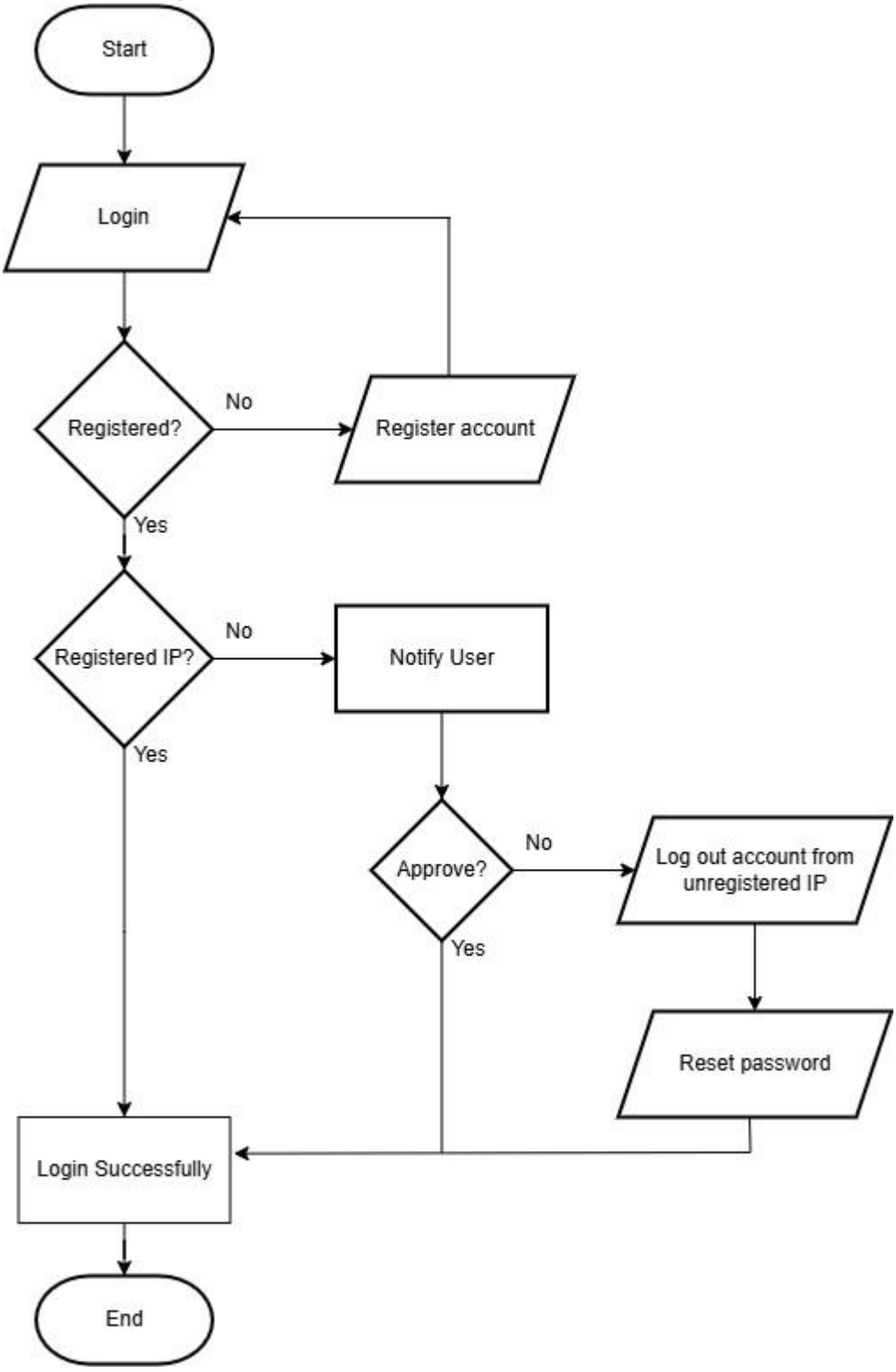


Figure 4.2: Flowchart Diagram

Figure 4.2 shows the flowchart of the proposed system. The flowchart outlines a secure authentication process designed to detect and respond to unauthorized login attempts effectively. The process begins when a user attempts to log in. The system first checks if the user has a registered account. If the account is not registered, the user is prompted to create one. If the account exists, the system then verifies whether the login is from a registered IP address. If the IP address is recognized, the user is granted access, and login to the system.

For unregistered IPs, the system sends a notification to the user about the suspicious login attempt. The user must then decide whether to approve or reject the login. If approved, the login is permitted, and login to the system. If rejected, the user can manually log out the account from the unregistered IP and prompt the user to reset their password as an added security measure.

4.1.2 Use Case Diagram

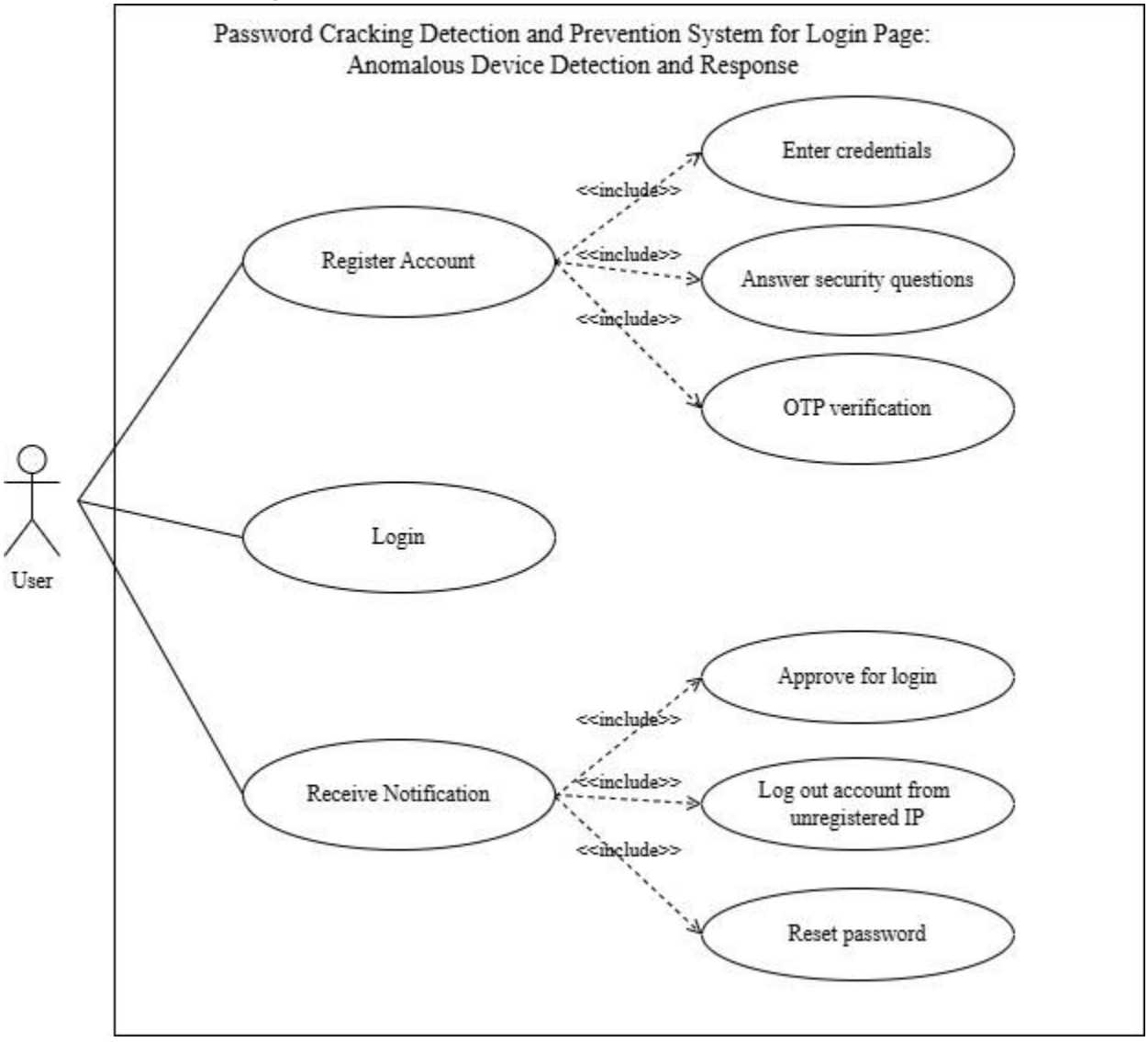


Figure 4.3: Use Case Diagram

Figure 4.3 shows the use case diagram of the proposed system. The main actor is the User, interacts with the system to perform primary actions such as account registration, and login. During the registration process, the user must complete requirement steps, including entering credentials, answering security questions, and performing OTP (One-Time Password) verification, in order to ensure a multi-factor authentication of the process. For login, the system verifies the credentials of the user while monitoring for suspicious activities. If an account is accessed from an unregistered IP address or device, the system then detects the anomaly and notifies the user,

whether to approve or reject the login. If rejected, the user can manually log out of the account and prompt the user to reset their password as a preventive measure.

4.1.3 Activity Diagram

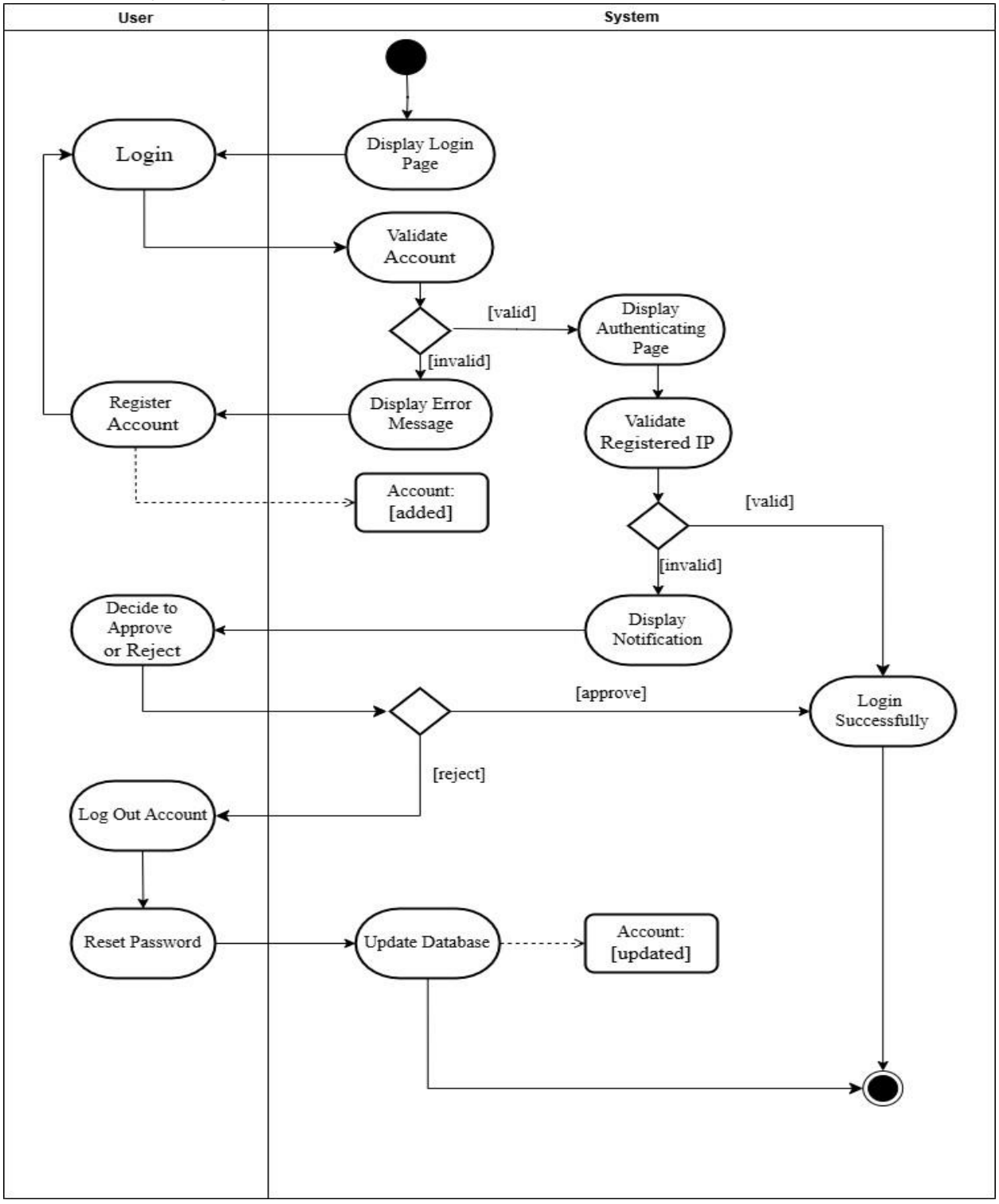


Figure 4.4: Activity Diagram

Figure 4.4 is the use case diagram of the proposed system. There are several processes in this process flow. First is the login and registration process. The system will be displaying the login page for users to prompt their credentials. If the account is validated, the user will proceed to the authenticating page waiting for authenticate. If not validated or registered, the system will bring the user to the registration page for the registration of a new user account. After registering an account, the registered account will be added to the database.

The second process is the account management process. If the login is from a registered IP address, it will let him log in successfully. If not, the system will send out a notification to the user to ask the user to make a decision. If the user decides to approve, then it will log in successfully. Else, the user can manually log out the account from the unregistered IP and reset their password as a preventive measure. The system will update the database of the user password.

4.1.4 Sequence Diagram

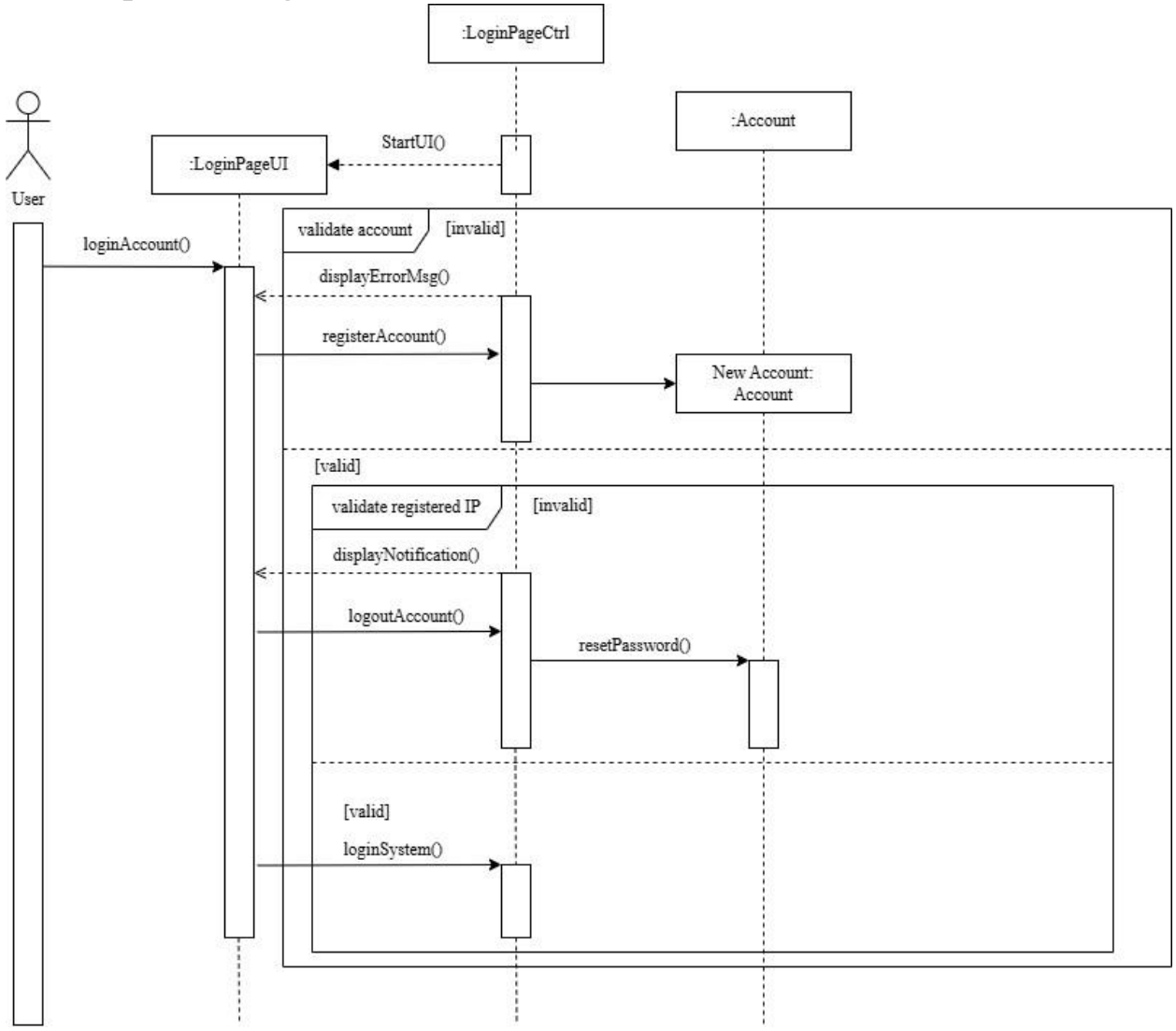


Figure 4.5: Sequence Diagram

Figure 4.5 shows the sequence diagram of the proposed system, illustrating the interactions between the User, LoginPageUI, LoginPageCtrl, and the Account system. The process begins with the user initiating a login request by invoking the loginAccount() method on the LoginPageUI, which then triggers the StartUI() function in the LoginPageCtrl to handle the request. The controller validates the user's credentials by calling the validate account() method on the Account system. If the credentials are invalid, an error message (displayErrorMsg()) is displayed to inform the user.

In cases where the user opts to register a new account, the LoginPageCtrl calls the registerAccount() method, creating a new account instance (New Account: Account) to store user information. For valid accounts, the system further validates the registered IP address through the validate registered IP() method. If the IP address is invalid, a security notification (displayNotification()) is sent to alert the user. Users can also manage their account by logging out using the logoutAccount() function or resetting their password with the resetPassword() method if needed.

Finally, if the credentials and IP validation are both successful, the system grants access to the user by executing the loginSystem() method. The diagram effectively represents key scenarios such as login attempts, account registration, password recovery, logout, and security notifications. It highlights the logical flow of operations, conditional checks, and the division of responsibilities between the user interface, controller, and account system.

4.2 Database Design

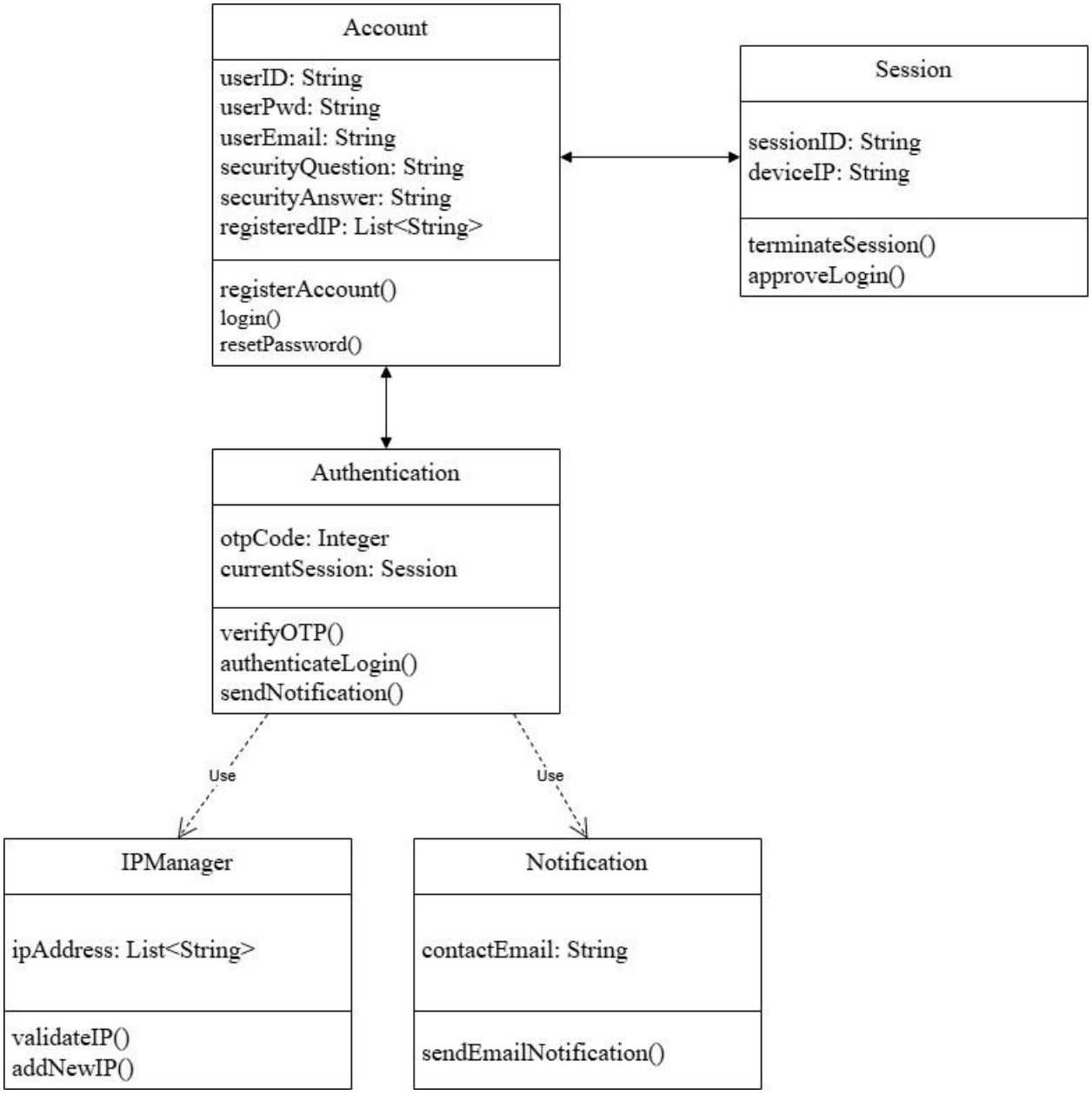


Figure 4.6: Class Diagram

The class diagram represents the design of a secure account management system that is aimed at enhancing user authentication and preventing unauthorized access. It features five interconnected classes such as Account, Session, Authentication, IPManager, and Notification. Each with specific roles to implement functionalities and maintain system security.

The Account class stores user details, including ID, password, email, security question, and registered IP addresses, with methods for registering an account, logging in, and resetting passwords. The Session class tracks login sessions through attributes like session ID and device IP, with methods to approve or terminate sessions. These classes ensure users can securely manage their accounts and monitor device usage.

The Authentication class is central to multi-factor authentication, keeping track of attributes such as OTP codes and current sessions. It provides methods for verification of OTPs, authentication of login attempts, and sending notifications. The IPManager class ensures access by only authorized devices through validating and registering new IP addresses. Meanwhile, the Notification class handles the sending of user alerts via email regarding suspicious login attempts or requests for approval.

4.2.1 Choice of Database

MySQL will be used for the database management system for this proposed secure login system that shall employ MFA. This is because MySQL offers ease of use, scalability, high performance, and strong security features, hence rendering it an ideal candidate in the implementation of a secure login system using MFA. These attributes have combined well with the requirements of the proposed system for efficient and safe authentication of users.

MySQL's user-friendly interface and intuitive commands simplify the development process, enabling the focus to remain on implementing core MFA features rather than struggling with database complexities. This accessibility makes MySQL suitable for both novice and experienced developers, allowing for faster system deployment and maintenance.

The scalability of the proposed system is crucial, especially in cases when the number of users increase or login attempts reach a high level. MySQL provides excellent scalability options, including vertical scaling (hardware upgrades) and horizontal scaling (clustering and sharding).

This will make the system able to handle increased loads without a loss in performance and provide reliability for user authentication and session management.

The next benefit of MySQL is that it gives a great performance. It proposes some very important features that support performance like indexing and caching mechanisms. These support rapid OTP verification, IP validation, and response to users needs at runtime. It prevents most of the delay-related latency that impacts the user during their login or response.

Security is a significant demand for the proposed MFA system. MySQL provides enhanced security features, including the use of SSL for data encryption, granular access control, and advanced authentication mechanism, which come in handy in securing password information, security questions, and IP addresses. These features ensure that user accounts remain safeguarded against unauthorized access and potential data breaches.

Additionally, MySQL enjoys a large and active community that provides support, troubleshooting resources, and best practices for implementation. The comprehensive documentation goes further to detail the guidance on integrating and maintaining secure database systems for smooth operation throughout the system life cycle.

In conclusion, MySQL's combination of ease of use, scalability, performance, and security makes it an excellent choice for implementing the proposed secure login system with MFA. Thus, it provides for a reliable authentication and protection of user data using MySQL capabilities while giving a seamless experience, thus increasing trust and confidence in its users.

4.3 User Interface Design

The UI design serves as the first line of interaction between the user and the application at the front end. The success of any UI design lies in guaranteeing the smoothness and intuitiveness of the user experience. Therefore, it directly impacts users' satisfaction with the developed application. This project concerns designing the UI through the construction of wireframes and prototypes.

4.3.1 Wireframe

Wireframes are designed to layout the required content and functionality of the system for an improved user experience in the usage of the system.

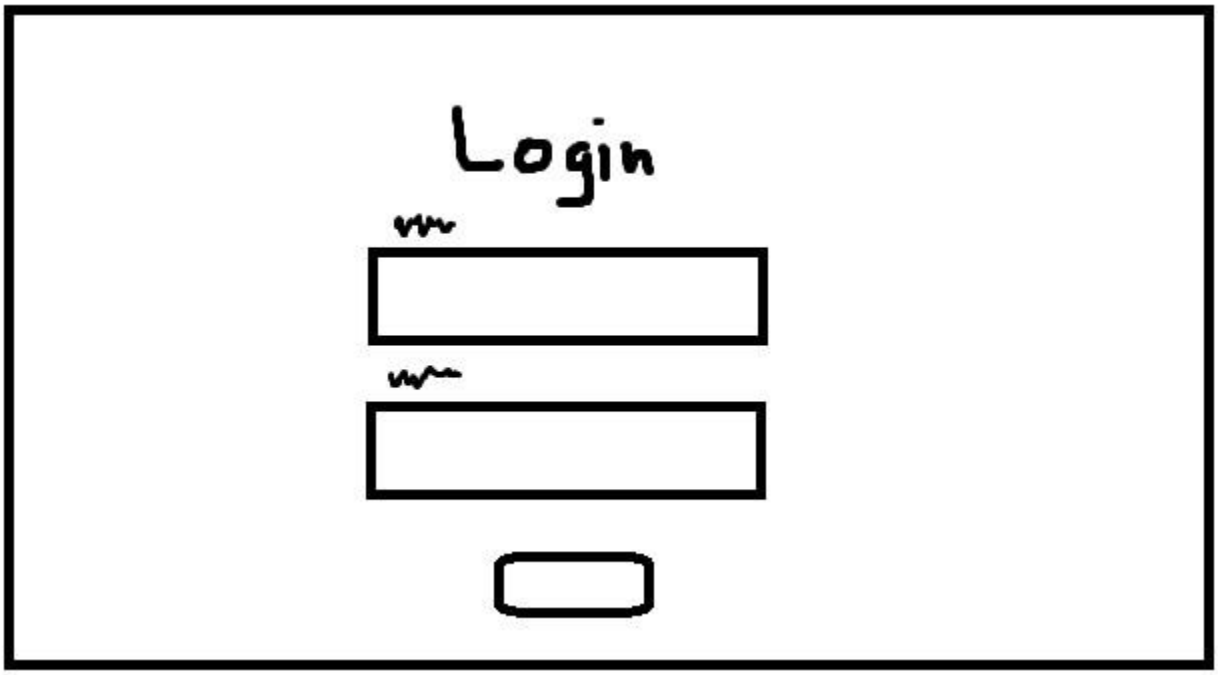


Figure 4.7: Layout of Login Page

Figure 4.7 shows the layout of the login page. The squiggly lines represent the placeholder text or labels displayed on the page. The two rectangular boxes are input fields where users enter their login credentials. The curvy box represents the button users click to proceed with the login process.

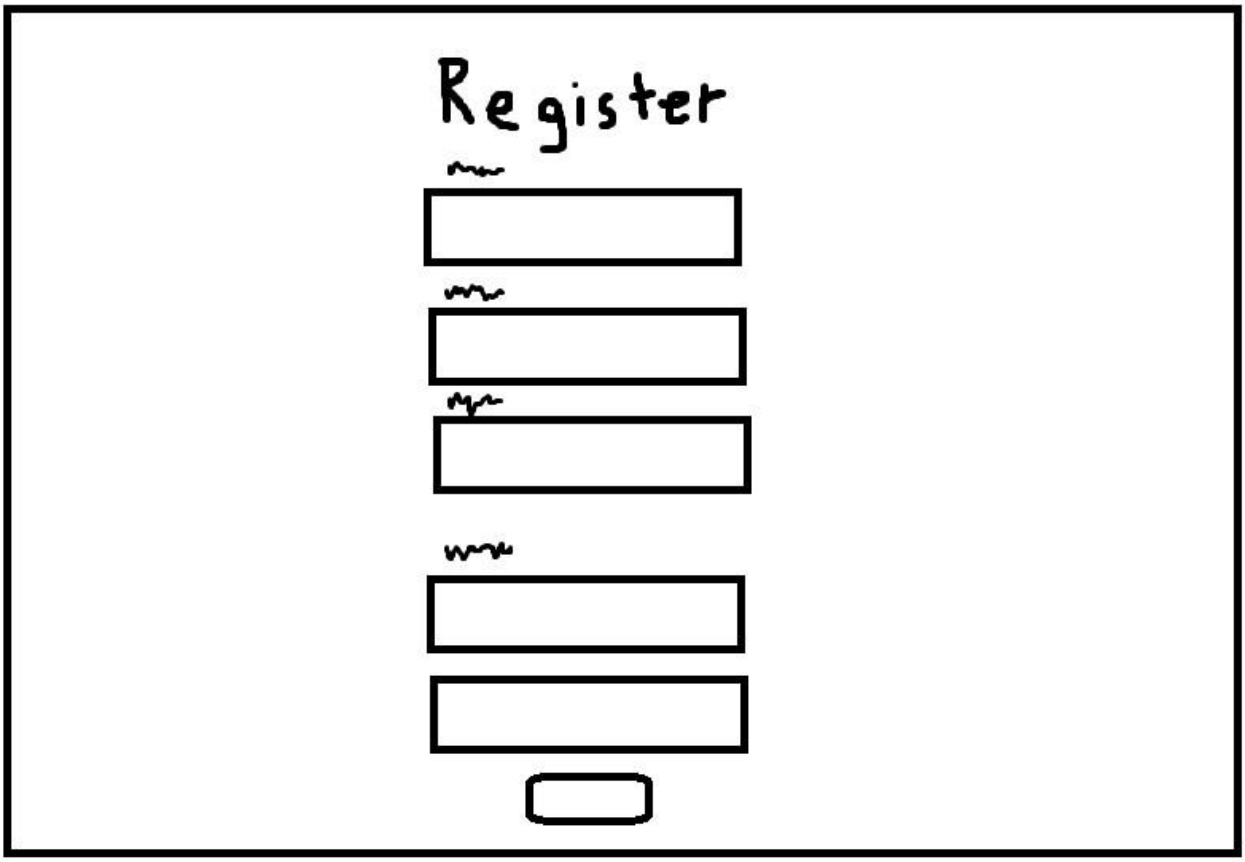


Figure 4.8: Layout of Registration Page

Figure 4.8 shows the layout of the registration page which shares a similar design with the login page. The squiggly lines represent placeholder text or labels. The rectangular boxes are input fields for user information, while the curvy box represents the button to proceed. Additionally, two rectangular boxes at the bottom are designated for users to answer the security questions.

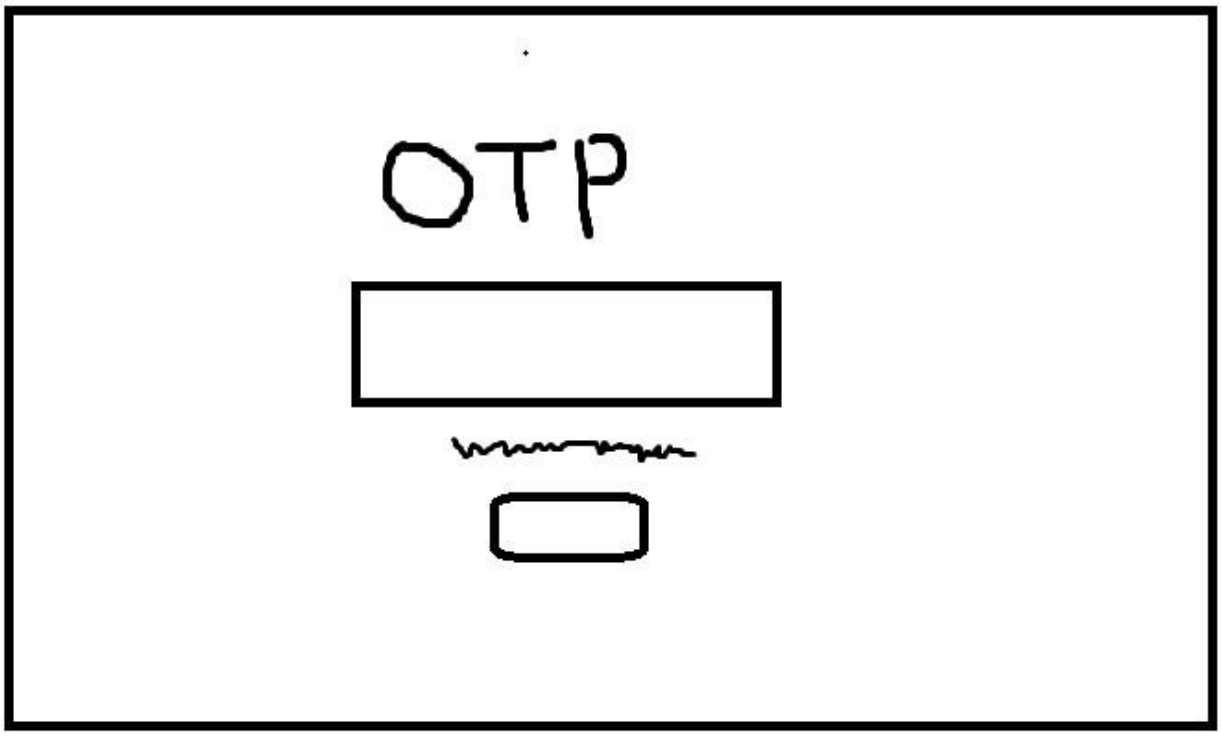


Figure 4.9: Layout of OTP Verification Page

Figure 4.9 shows the layout of the OTP verification page , which shares a similar design with the login and registration pages. This page allows users to enter the OTP for verification.



Figure 4.10: Layout of Authenticating Page

Figure 4.10 shows the layout of the authenticating page. This page design only squiggly lines, indicating that the login process is in progress.

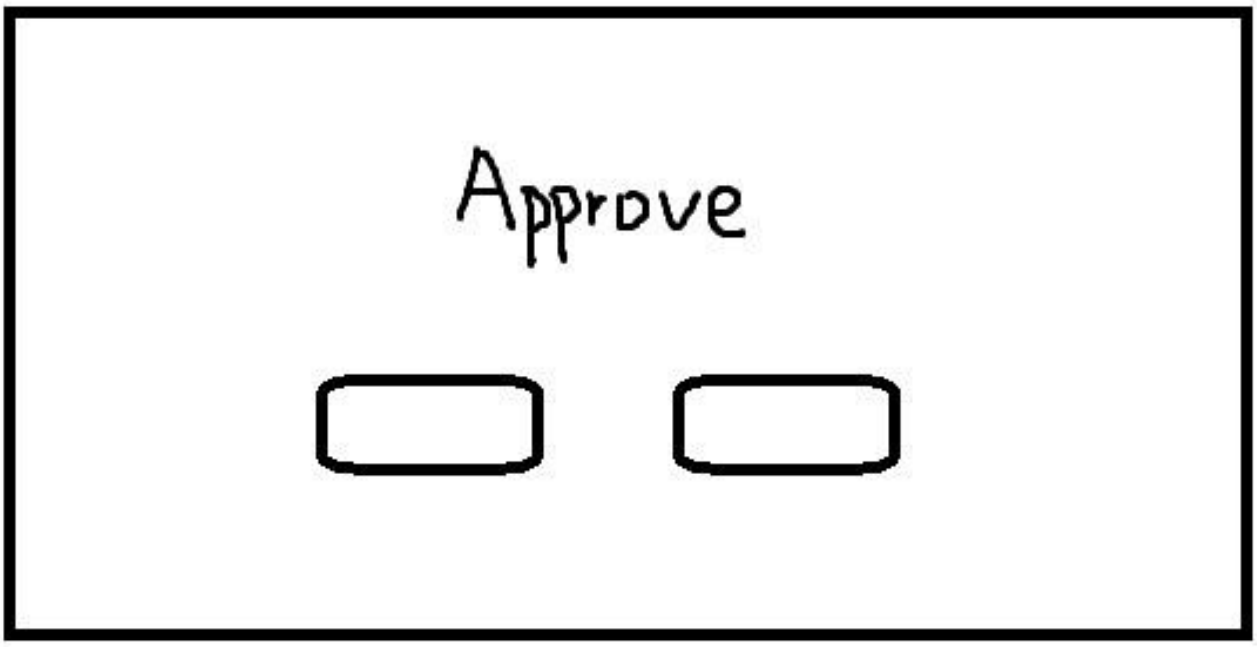


Figure 4.11: Layout of Approver Page

Figure 4.11 shows the layout for the approver page. This page only contains the question and two curvy box which are buttons, asking whether the user is logging in with a new device.

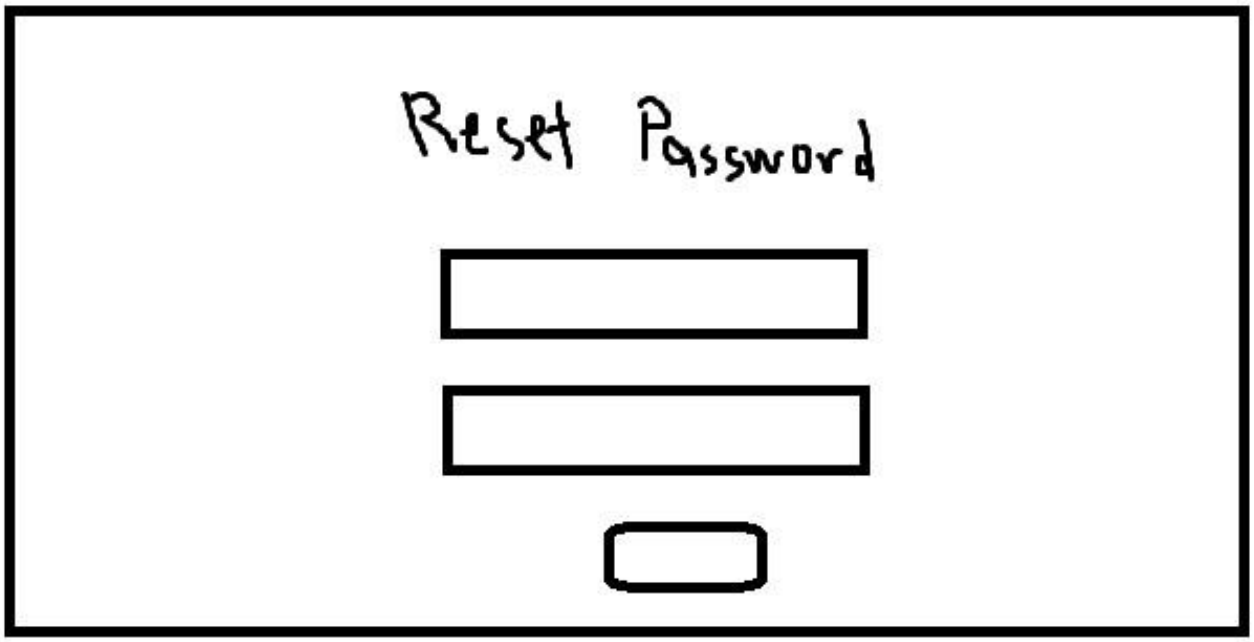


Figure 4.12: Layout of Reset Password Page

Figure 4.12 shows the layout for the reset password page. This page consists of two rectangle boxes which are the input field and one curvy box represents a button for the user to reset their password.

4.3.2 Prototype

Prototype design is used to test the practicability of the wireframe design. Below is the display of prototype design for each page based on the wireframe.



Figure 4.13: Prototype Design of Login Page

Figure 4.13 shows the prototype design of the login page. The interface focuses on crucial elements, including fields for email and password to ensure a minimal and distraction-free layout. Key user experience (UX) principles such as simplicity, consistency, and clear call-to-action buttons (e.g., “Login” and “Register”) guide users effortlessly through the process. Accessibility features are considered by ensuring readable fonts, adequate color contrast, and compatibility with screen readers for visually impaired users. Additionally, keyboard navigation is also supported to accommodate users with limited mobility.

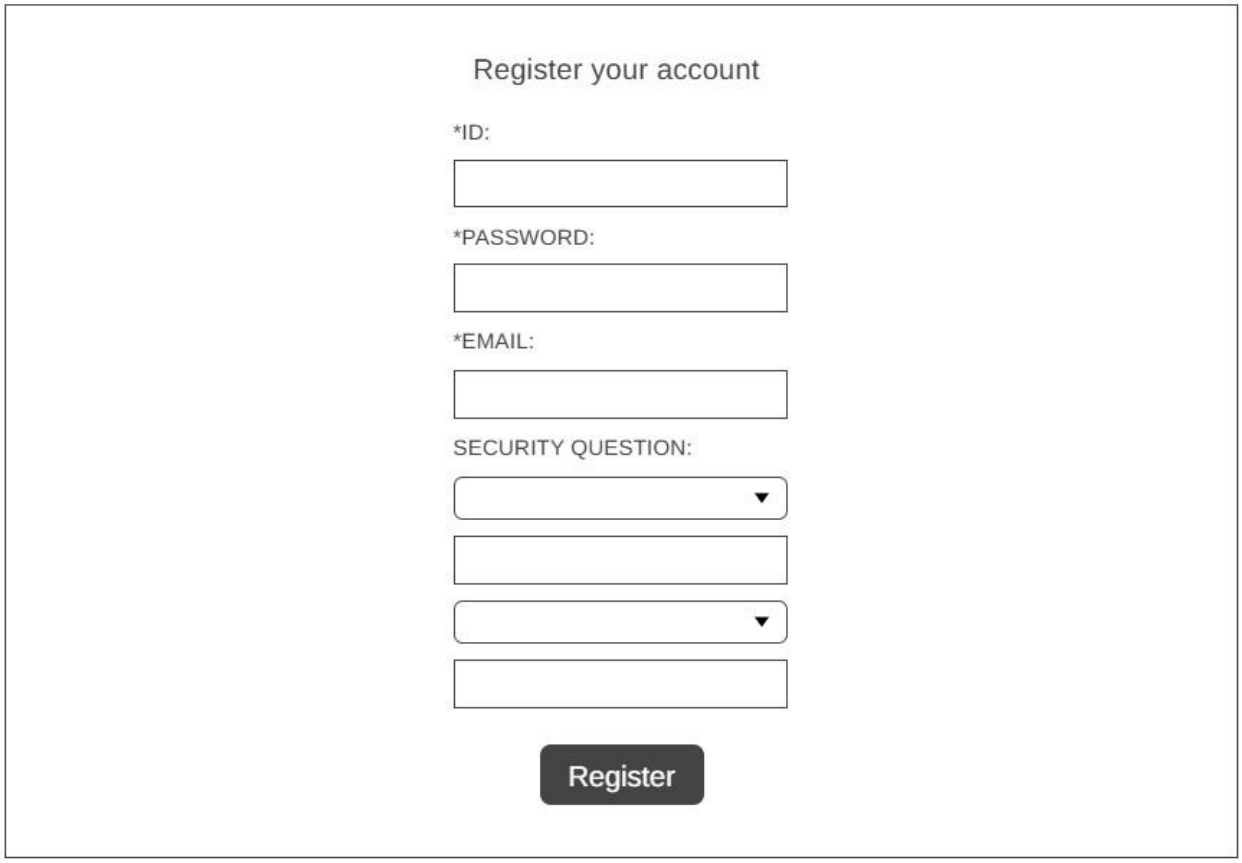


Figure 4.14: Prototype Design of Registration Page

Figure 4.14 shows the prototype design of the registration page. The design is clear and intuitive, starting with important fields such as ID, password, and email, followed by security questions to enhance account protection. Mandatory fields are marked with an asterisk (*) to guide users, minimizing confusion and ensuring that critical information is provided. Dropdown menus for security questions simplify user input, making the process more convenient. Accessibility considerations are incorporated to make the page usable by a diverse audience. Each input field is labeled clearly to support screen readers so that visually challenged users can fill out the forms effectively. The form also supports keyboard navigation, which is important for users with mobility impairments. Additionally, the text size and contrast are ideal to enable viewers with poor vision to navigate the form successfully.



Figure 4.15: Prototype Design of OTP Verification Page

Figure 4.15 shows the prototype design of the OTP verification page. In this page, the OTP verification is effectively applying user experience principles through simplicity and intuition. The clear instructions, "Enter the OTP sent to example@gmail.com," tells the user what they are being asked to do, while the "Resend" option tries to solve some problems related to delayed or missing OTP delivery. The "Verify" button is large and clear, so users can easily identify what they need to do to complete the process. Accessibility considerations include short and direct language, a clean layout, and adequate space for input fields to accommodate all types of users.

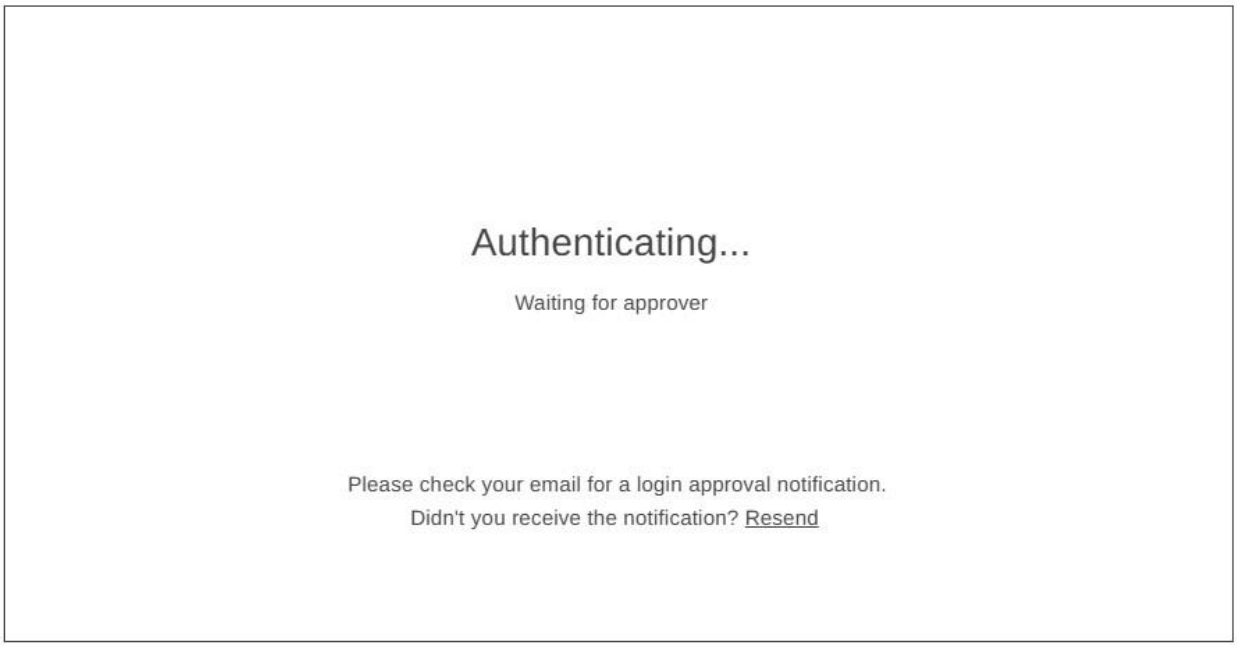


Figure 4.16: Prototype Design of Authenticating Page

Figure 4.16 shows the prototype design of the authenticating page. This page effectively adheres to UX principles by providing clear feedback to the user during the login process. The message "Authenticating... Waiting for approver" tells the users what is happening, and the instruction to check their email for a notification explains what happens next. The "Resend" option is convenient if there are some delays or the notification has not been seen. Accessibility is considered by using simple and direct language so that users can easily follow the process.

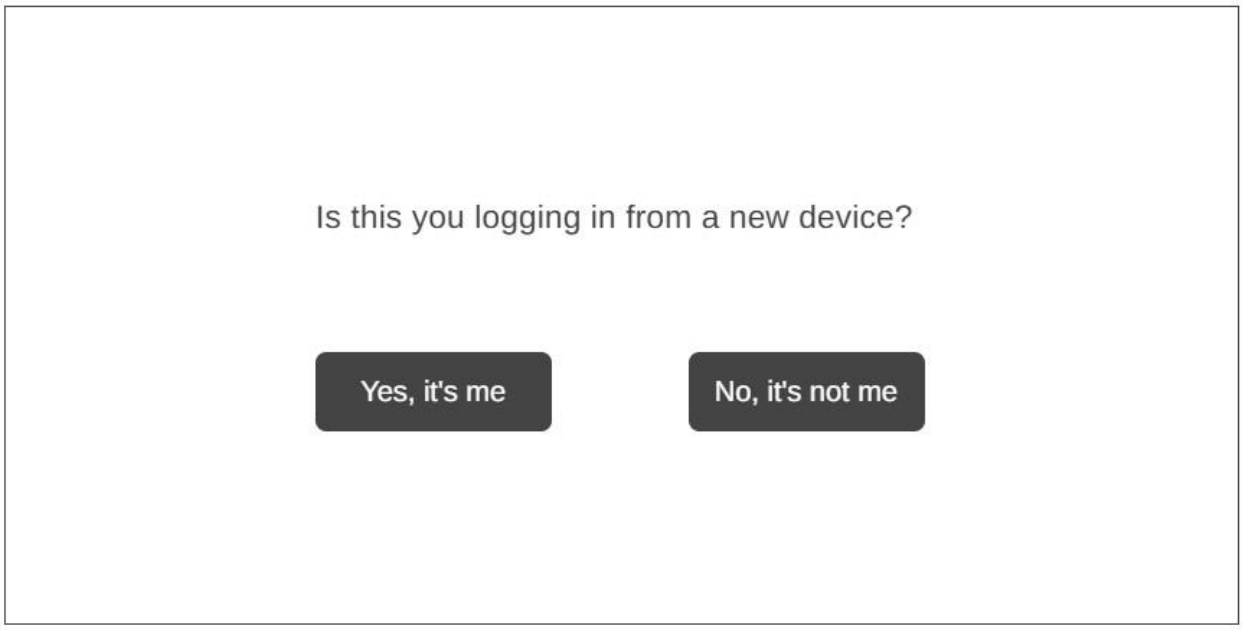


Figure 4.17: Prototype Design of Approver Page

Figure 4.17 shows the prototype design for the approver page. The user after receiving the notification, will provide a link to bring the user to this page. The simple and clear interface ensures ease of use, providing a straightforward question with two highly distinguishable options: "Yes, it's me" and "No, it's not me." This reduces cognitive load, allowing users to make quick decisions without confusion. The large and well-labeled buttons are accessible for users with motor impairments because they are easy to click both on desktop and mobile devices. Additionally, the high contrast between the text and button backgrounds is readable for users with visual impairments, aligning with accessibility standards like WCAG. It designates an immediate response option so that through the design, the user can quickly confirm or deny suspicious login activity quickly. This improves security and builds trust and user satisfaction. This focus on usability and accessibility ensures all users will have a seamless experience, regardless of ability or circumstance.

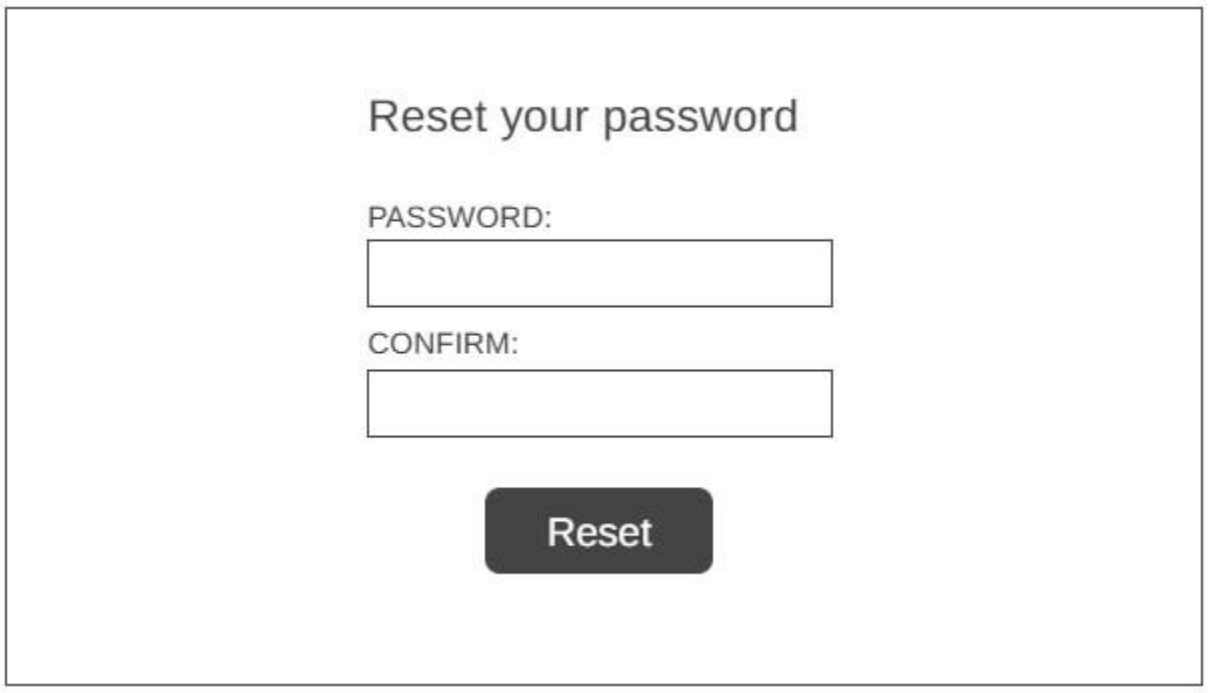


Figure 4.18: Prototype Design of Reset Password Page

Figure 4.18 shows the prototype design for the reset password page. Once the user clicks “No it’s not me” at the approver page, it will ask the user to reset their password. The clean design with clearly labeled fields like "Password" and "Confirm", will reduce user confusion and make the task intuitive, ensuring that users can complete the process quickly and accurately. The "Reset" button will appear with high contrast and well-sized, making it accessible for users with motor impairments using an assistive technologies input method. The consistent and minimalistic layout ensures a smooth experience on both desktop and mobile devices, catering to diverse user needs.

4.4 Detailed Design

The detailed design of the system encompasses several components and modules combined together for a specific purpose that adds to functionality and security. The Account Management Module shall deal with user account operations such as registration, login, and reset password. It stores user credentials, email addresses, and answers to security questions securely, and provides APIs to help in account registration and password reset processes. The Authentication Module

implements MFA via OTP verification to ensure that only authorized users have access to the system. It includes APIs related to the generation and validation of OTPs, adding an additional layer of security to the login.

The Session Management Module tracks the active user sessions, monitors login activity, and detects unauthorized access attempts, and notifies users in cases of anomalies, such as logins from unfamiliar IP addresses. It provides APIs for starting and terminating user sessions, making the system more secure and transparent to the users. The IP Management Module further reinforces security in the validation and registration of IP addresses, ensuring that login attempts are only made from authorized devices. It includes APIs for checking the validity of IP addresses and adding new ones to the list of trusted devices. Besides, the Notification Module supports the users through messages and notifications for any occurrence of an event, which relates to security features, such as notifications of login from unregistered IPs or password changes in accounts.

The design decisions aim at offering the highest possible degree of security with ease in operation. It strengthens the MFA authentication in a way that verification must be two-fold, via password and an OTP. Although MFA increases the effort required for users during login, the added security justifies this trade-off. Similarly, IP Validation adds another layer of protection by restricting logins to registered IP addresses. While this may cause slight delays when users log in from new devices, the benefit of preventing unauthorized access outweighs the inconvenience. For Session Monitoring, which offers further security by tracking the activity of the users and alerting them immediately upon detecting suspicious behavior. This takes additional resources on the database, but the added security makes it well worth the added resource usage.

In the design of the system, MySQL was chosen for its scalability, ease of use, and robust security features. While NoSQL databases may offer better performance in high-scale read operations, MySQL's ability to handle complex queries and maintain relational data integrity better fits the needs of the system. Additionally, a lot of effort was put into designing a simple and accessible user interface. This ensures that the system is intuitive and user-friendly for a diverse audience,

though it slightly extended the development timeline. The balance of security, usability, and scalability was central in the design decisions to ensure that the system is able to handle an increasing number of users while maintaining its functionality and ease of use.

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